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Program Project Report on the Topic:
Automated Product Description Generation From Images
(interim, the first stage)

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Abstract

We develop a lightweight prototype that generates a short product description from a single product image. This is motivated by the scale of e-commerce catalogs, where items and photos are updated frequently and manual description writing becomes costly and inconsistent. Our approach combines a pre-trained vision - language model to extract information supported by the image and a language model to convert these cues into a concise description. Two questions guide the work:

1. Is the text actually useful on a product page?
2. What typical failure modes appear - for example, hallucinated details or overly generic phrasing?

The output is evaluated using standard captioning metrics and supported by a brief manual check to verify that stated attributes are grounded in the image.

Аннотация

В работе разрабатывается прототип, который по одной фотографии формирует короткое описание товара. Задача актуальна для онлайн-ритейла: каталоги велики, ассортимент быстро обновляется, а ручное написание текстов плохо масштабируется и приводит к разнородному стилю. Подход основан на использовании предварительно обученной vision - language модели для извлечения визуально подтверждаемых признаков и языковой модели для генерации компактного и понятного описания. В фокусе проекта - практическая полезность текста и типовые ошибки, в частности приписывание несуществующих деталей и слишком общие формулировки. Качество оценивается стандартными метриками image captioning и дополнительной ручной проверкой соответствия описания изображению.

1 Introduction

Good product descriptions help shoppers grasp what they’re looking at, improve search and filters, and cut down on “What is this?” questions. The main challenge is scale: catalogs grow, photos get refreshed, and many similar items appear, so text drifts out of date fast.

At the same time, image-to-text models have leapt forward. Classic captioning already writes fluent sentences, and newer vision - language systems add richer context. That makes them tempting for product copy.

But store copy isn’t just “describe the scene.” It has to stick to the item, call out attributes people care about, and stay inside what’s actually visible. If the model guesses the fabric, invents a brand, or tacks on a feature, this reduces user trust. So the goal is clear: generate text that stays grounded and consistent in an e-commerce setting.

This project sketches and tests a prototype pipeline that converts a product image into a short description, then shows where it works and where it fails.

1.1 Project objectives and planned results

The primary objective of this project is to design and implement a prototype service capable of generating textual product descriptions from images. The expected result is a working pipeline that accepts product images as input and produces readable and informative descriptions as output. The project also aims to evaluate the quality of generated descriptions and identify limitations of image-based description generation in the e-commerce domain.

2 Literature review

Sharma, A., Singh, H., Pant, M. (2025) - Pixels to prose: A comprehensive survey of image captioning techniques

Description. This paper presents a comprehensive survey of image captioning methods, covering the evolution of techniques from early template-based and retrieval-based approaches to modern deep learning models, including transformers and vision - language architectures. The authors systematically analyze supervised, unsupervised, and reinforcement learning approaches, as well as commonly used datasets and evaluation metrics.

Connection to the topic. This survey is directly relevant to the present project, as it provides a structured overview of existing image captioning techniques that form the foundation for automatic product description generation. It helps position the project within current research and supports the choice of modern multimodal models for generating textual descriptions from images in e-commerce settings.

Methods and models. The paper reviews encoder - decoder architectures based on CNNs and RNNs, attention-based models, transformer-based approaches, and emerging vision - language models. It also discusses commonly used datasets such as MS COCO and Flickr, along with evaluation metrics including BLEU, CIDEr, METEOR, and SPICE.

Results. The authors conclude that transformer-based and multimodal architectures significantly outperform earlier methods in terms of caption quality and semantic alignment. They also highlight remaining challenges, such as domain adaptation and evaluation limitations, which are particularly relevant for specialized applications like product description generation.

Abdulgalil, H.D., Basir, O.A. (2025) - Next-generation image captioning: A survey of methodologies and emerging challenges

Description. This article surveys recent advances in image captioning with a focus on transformer-based models and Multimodal Large Language Models (MLLMs). The authors analyze how image captioning systems have evolved from traditional encoder - decoder architectures to large-scale multimodal models capable of joint visual and linguistic reasoning.

Connection to the topic. The paper is highly relevant to this project as it discusses state-of-the-art multimodal models that align closely with the goal of generating meaningful product descriptions from images. In particular, the transition toward MLLMs directly supports the use of large language models for flexible and domain-adaptive text generation in e-commerce applications.

Methods and models. The authors review CNN - RNN architectures, attention-based models, transformer encoders and decoders, and recent MLLMs such as CLIP-based and LLaVA-like systems. The paper also compares different visual feature representations and fusion strategies used in modern captioning pipelines.

Results. The survey demonstrates that transformer-based and multimodal models achieve superior performance on standard benchmarks such as MS COCO. At the same time, the authors identify open challenges, including hallucination, grounding, and computational cost, which are

important considerations when designing practical product description generation systems.

Parvin, H., Naghsh-Nilchi, A.R., Mohammadi, H.M. (2023) - Transformer-based local-global guidance for image captioning

Description. This paper proposes a transformer-based image captioning model that introduces both local and global guidance mechanisms to improve word selection during caption generation. The authors aim to address limitations of standard transformer models that fail to adequately capture intra-object relationships and complex visual dependencies.

Connection to the topic. The proposed approach is relevant to this project because product images often contain multiple visual attributes that must be described consistently. The idea of combining local and global guidance can help generate more accurate and detailed product descriptions, especially for visually complex items.

Methods and models. The model consists of a transformer-based generator network for word-level prediction and a selector network that evaluates candidate words using a shared image - text embedding space. The approach eliminates recurrence and convolution, enabling parallel processing and improved efficiency. Experiments are conducted on MS COCO and Flickr30k datasets.

Results. The experimental results show that the proposed method outperforms several state-of-the-art models across standard evaluation metrics. The authors demonstrate improved semantic consistency and caption quality, indicating the effectiveness of combining local and global visual guidance.

Wang, R., Wu, Y., Sheng, Z. (2025) - ClipCap++: An efficient image captioning approach via image encoder optimization and LLM fine-tuning

Description. This paper introduces ClipCap++, an enhanced version of the ClipCap image captioning model that integrates domain-specific knowledge into large language models. The authors focus on improving caption quality and generalization, particularly in specialized or low-resource domains.

Connection to the topic. ClipCap++ is especially relevant for this project because it directly addresses domain adaptation, which is critical for e-commerce product descriptions. The approach demonstrates how pre-trained vision - language models and LLMs can be adapted to generate more accurate descriptions for specific product categories.

Methods and models. The model builds on CLIP for visual encoding and GPT-based language models for text generation. It introduces key - value pair modules for few-shot learning and residual connections to balance pre-trained and domain-specific features. Both hard and soft visual prompts are used to guide the language model during caption generation.

Results. Experimental results show that ClipCap++ outperforms the original ClipCap and other baselines in accuracy, efficiency, and cross-domain generalization. The authors highlight significant reductions in training time while maintaining high caption quality, making the approach suitable for practical deployment.

Elbedwehy, S., Medhat, T., Hamza, T., Alrahmawy, M.F. (2022) - Efficient image captioning based on vision transformer models

Description. This paper investigates the use of different Vision Transformer (ViT) architectures as visual encoders in image captioning systems. The authors systematically evaluate several transformer-based vision models, including DINO, PVT, XCiT, and Swin Transformer, focusing on their impact on caption quality and model efficiency.

Connection to the topic. The study is relevant to this project as it provides empirical evidence on how the choice of visual encoder affects caption generation quality. For e-commerce product images, selecting an efficient and expressive visual backbone is critical for accurately capturing product attributes such as shape, texture, and structure.

Methods and models. The proposed captioning framework follows an encoder - decoder architecture, where different ViT variants are used for visual feature extraction and combined with a language decoder. The models are evaluated using standard captioning benchmarks and metrics, including BLEU and CIDEr, and compared against CNN-based baselines.

Results. The results show that Swin Transformer-based encoders outperform other tested architectures in terms of caption accuracy and semantic consistency. The authors conclude that hierarchical vision transformers provide a strong balance between performance and computational efficiency, making them suitable for practical captioning applications.

Jung, Y., Cho, I., Hsu, S.-H., Golparvar-Fard, M. (2024) - VisualSiteDiary: Vision - language transformers for domain-specific captioning

Description. This paper presents VisualSiteDiary, a vision - language transformer model designed for generating human-readable captions for construction site photologs. The authors introduce a new dataset and focus on producing detailed, context-aware captions tailored to a specific application domain.

Connection to the topic. Although the target domain differs from e-commerce, this work is highly relevant as it demonstrates effective domain adaptation of image captioning models. The idea of tailoring caption generation to domain-specific requirements directly aligns with the goal of generating accurate and informative product descriptions.

Methods and models. VisualSiteDiary is based on a Vision Transformer backbone combined with a pretrained vision - language model. The system incorporates pseudo-region features, domain-specific pretraining, and fine-tuning strategies to improve caption relevance and retrieval performance.

Results. Experimental evaluations show that VisualSiteDiary outperforms state-of-the-art captioning models on both caption quality and image retrieval tasks. The results highlight the importance of domain-specific data and adaptation strategies for improving caption usefulness in real-world applications.

Song, L., Wang, Y., Shi, L., Yu, J., Li, F., Xiang, S. (2025) - Transformer with token attention and attribute prediction for image captioning

Description. This paper proposes an image captioning framework that reduces compu-

tational complexity by selecting informative visual tokens and explicitly predicting semantic attributes. The authors aim to address inefficiencies caused by processing redundant visual tokens in Vision Transformers.

Connection to the topic. The approach is relevant for scalable e-commerce systems, where efficiency and speed are important. Reducing unnecessary computation while preserving caption quality is beneficial for generating product descriptions at scale.

Methods and models. The model introduces a token attention module that prunes uninformative image patches and an attribute prediction module that strengthens the alignment between visual features and language generation. The architecture follows an encoder - decoder transformer framework and is evaluated on MS COCO and NoCaps datasets.

Results. The experimental results demonstrate that the proposed method significantly reduces computational cost while maintaining competitive captioning performance. The authors report large reductions in FLOPs with minimal loss in caption accuracy, indicating the effectiveness of token-level optimization.

Balachandran, A., Masum, M. (2025) - A multimodal framework for enhancing e-commerce information management

Description. This paper introduces BiLens, a multimodal framework that integrates image captioning, vision transformers, and large language models to improve product search and recommendation in e-commerce systems. The framework combines visual and textual representations to better capture user intent.

Connection to the topic. This work is directly connected to the present project, as it demonstrates a practical application of image captioning in e-commerce. The use of generated captions to enhance downstream tasks supports the motivation for automatic product description generation.

Methods and models. BiLens employs vision transformers for visual feature extraction and large language models for caption generation. The generated captions are converted into embeddings and fused with visual representations using an early fusion strategy. The framework is evaluated on real-world product datasets.

Results. The results show that multimodal fusion significantly outperforms image-only and text-only baselines in recommendation accuracy. The study confirms that high-quality image captions contribute to more accurate, context-aware e-commerce systems.

Memiş, A., Yıldız, S. (2025) - Tiny TR-CAP: A novel small-scale benchmark dataset for image captioning

Description. This paper introduces Tiny TR-CAP, a small-scale benchmark dataset for image captioning consisting of 1,076 images with five human-written captions per image. The dataset was originally created in Turkish and later translated into English, allowing evaluation of captioning models in a low-resource setting.

Connection to the topic. This work is relevant to the present project because real-world e-commerce applications often lack large, high-quality labeled datasets. Tiny TR-CAP demon-

strates that meaningful evaluation and comparison of image captioning models is possible even with limited data, which aligns with the constraints of practical product description generation.

Methods and models. The authors evaluate a wide range of modern captioning systems, including transformer-based models and multimodal large language models such as BLIP, BLIP-2, CLIPCap, MiniGPT-4, and LLaVA. Standard automatic metrics such as BLEU, METEOR, ROUGE, CIDEr, and SPICE are used for evaluation.

Results. The experimental results show that modern transformer-based and vision - language models maintain reasonable performance even on small datasets. The study highlights differences in robustness across models and emphasizes the importance of careful evaluation when data availability is limited.

Chen, J., Li, X. (2024) - Dual-adaptive interactive transformer for image captioning

Description. This paper proposes the Dual-Adaptive Interactive Transformer (DAIT), an image captioning architecture designed to improve cross-modal interaction between visual and textual features. The model aims to address limitations of static feature fusion in existing captioning systems.

Connection to the topic. For e-commerce product descriptions, accurate alignment between visual content and generated text is essential. DAIT is relevant to this project as it explicitly models adaptive interaction between modalities, which can help reduce mismatches between visual attributes and textual descriptions.

Methods and models. DAIT consists of an Adaptive Interactive Encoder that enhances visual and textual features using gating and normalization mechanisms, and an Adaptive Interactive Decoder that dynamically balances the contribution of visual and textual information during caption generation. The model is evaluated on standard benchmarks such as MS COCO.

Results. The authors report consistent improvements over baseline and state-of-the-art models across multiple captioning metrics. The results indicate that adaptive multimodal interaction leads to better semantic consistency and more accurate captions.

Thobhani, A., Shah, R., Patel, D. (2025) - Visual attention-based refined visual features for image captioning

Description. This paper addresses the limitation of static visual representations in image captioning models by introducing a framework that dynamically refines visual features during the caption generation process. The goal is to improve the relevance of visual features at each decoding step.

Connection to the topic. Product descriptions often require the model to focus on different attributes at different stages of generation. The dynamic refinement strategy proposed in this work aligns with the need to sequentially attend to relevant product features such as category, color, and details.

Methods and models. The framework includes a Visual Weighting Module (VWM) that reweights visual features based on the partial caption, and an Enhanced Feature Attention Module (EFAM) that further refines the attended features. The approach is integrated into a

transformer-based captioning architecture.

Results. Experimental evaluations demonstrate improved performance compared to baseline models on standard captioning datasets. The authors show that dynamically refined visual features lead to more accurate and context-aware captions.

Li, W., Li, X., Li, Z., Yu, J., Chen, P. (2025) - HI-Captioner: End-to-end image captioning based on hierarchical multi-scale encoding

Description. This paper presents HI-Captioner, an end-to-end image captioning framework that leverages hierarchical multi-scale visual encoding and cross-modal interactive decoding. The model aims to improve both fine-grained detail extraction and global semantic coherence.

Connection to the topic. The ability to capture multi-scale visual information is particularly important for product images, where both overall appearance and small details matter. HI-Captioner provides architectural insights that are relevant for generating detailed and accurate product descriptions.

Methods and models. The proposed framework uses a Swin Transformer-based visual encoder with hierarchical positional encoding and a cross-modal interactive decoder that strengthens alignment between visual and textual representations. Ablation studies are conducted to analyze the contribution of individual components.

Results. The authors report improved performance over baseline models on MS COCO and Flickr datasets. The results confirm that hierarchical multi-scale encoding and enhanced cross-modal interaction contribute to more detailed and coherent captions.

Summary. Across the papers we read, the trend is unmistakable: transformer-based, multimodal models dominate. They write smoother prose and align image and text better than older encoder - decoder setups, yet most results come from broad captioning datasets, not product-centric tasks.

For catalog copy, sounding natural isn't enough - the text has to cling to visible attributes and avoid confident guesses. Plugging that gap is why this project exists: we apply modern captioning approaches in the product domain and judge them on reliability and consistency, not just benchmark scores.

3 Methodology

3.1 Implementation snapshot: caption generation module

The prototype captioner is implemented as a small wrapper around a pre-trained BLIP captioning model. The wrapper supports an optional LoRA adapter (PEFT) and uses mixed precision on GPU/MPS to keep inference lightweight.

```
1 from __future__ import annotations
2 import torch
3 from PIL import Image
4 from transformers import BlipProcessor, BlipForConditionalGeneration
5 from .config import pick_device
6 from peft import PeftModel
7
8 class BlipCaptioner:
9     def __init__(self, model_name: str, adapter_path: str | None =
None):
10         self.model_name = model_name
11         self.device, self.amp_dtype = pick_device()
12
13         self.processor = BlipProcessor.from_pretrained(model_name)
14         self.model =
BlipForConditionalGeneration.from_pretrained(model_name,
use_safetensors=True)
15         if adapter_path:
16             self.model = PeftModel.from_pretrained(self.model,
adapter_path)
17             self.model.to(self.device)
18             self.model.eval()
19
20     @torch.no_grad()
21     def caption(
22         self,
23         image_path: str,
24         max_new_tokens: int = 40,
25         num_beams: int = 3,
26     ) -> str:
27         image = Image.open(image_path).convert("RGB")
28         inputs = self.processor(images=image, return_tensors="pt")
29         inputs = {k: v.to(self.device) for k, v in inputs.items()}
30
31         use_amp = self.device in {"cuda", "mps"} and self.amp_dtype ==
torch.float16
```

```

32         if use_amp:
33             with torch.autocast(device_type=self.device,
dtype=self.amp_dtype):
34                 out = self.model.generate(
35                     **inputs,
36                     max_new_tokens=max_new_tokens,
37                     num_beams=num_beams,
38                 )
39         else:
40             out = self.model.generate(
41                 **inputs,
42                 max_new_tokens=max_new_tokens,
43                 num_beams=num_beams,
44             )
45
46         return self.processor.decode(out[0],
skip_special_tokens=True).strip()

```

1: Caption generation wrapper with optional LoRA adapter and AMP

3.2 Problem formulation

We need to turn a product image into a short description. The text should name the item and, when the photo makes it possible, mention basic attributes - colour, shape, or other category-specific details - while steering clear of anything the image doesn't show. In short: useful catalog copy, not creative storytelling. Our evaluation therefore checks how firmly the text stays tied to the photo.

3.3 Data description

We base the current prototype on the Fashion Product Images Dataset hosted on Kaggle. The collection includes photos of clothing, footwear, and related accessories, each accompanied by structured metadata such as category labels and basic product details.

In our setup, the images supply the visual input, while the accompanying titles or short descriptions act as reference text during evaluation. We split the data into training, validation, and test sets to prevent any information from the final test portion leaking into development.

Fashion is a deliberate starting point: garments come with clear visual cues—colour, garment type, silhouette—making the domain well suited for controlled experiments in image-driven description generation. The trade-off is limited generalisation. In later stages, we plan to broaden or swap the dataset to include other product categories and check whether the pipeline still produces reliable descriptions outside fashion.

3.4 System architecture

The pipeline has three pieces:

1. **Visual encoder** - digests the image and outputs a compact vector.
2. **Conditioning step** - reshapes that vector into something the language model can use (a learned projection or a structured prompt).
3. **Language model** - writes the final description.

Starting with pre-trained parts gives us a solid baseline fast. If early outputs feel bland or wrong, we can tweak prompts, adjust conditioning, or fine-tune lightly instead of retraining everything.

3.5 Evaluation protocol

We combine quantitative metrics with qualitative inspection. At the current baseline stage, we compute SacreBLEU and ROUGE (ROUGE-1, ROUGE-2, ROUGE-L, ROUGE-Lsum) against the reference text. These overlap-based metrics provide a simple quantitative signal for comparing model variants and prompt settings. While such metrics are not perfectly aligned with attribute precision in product descriptions, they offer a consistent baseline for evaluation.

In addition to automatic metrics, we manually inspect a subset of generated descriptions in order to identify typical failure modes, such as hallucinated attributes, incorrect product categories, repetitive phrasing, or fluent but overly generic text. A model may achieve acceptable overlap scores while still producing misleading descriptions, therefore qualitative review remains essential.

Implementation snapshot: automatic evaluation

The following code excerpt illustrates how automatic metrics are computed and how a small sample of predictions is exported for manual inspection.

```
1 def compute_metrics(preds: List[str], refs: List[str]) -> Dict[str,
2     Any]:
3     # BLEU
4     bleu = evaluate.load("sacrebleu")
5     bleu_res = bleu.compute(predictions=preds, references=[[r] for r
6     in refs])
7     # ROUGE (rouge1/rouge2/rougeL/rougeLsum)
8     rouge = evaluate.load("rouge")
9     rouge_res = rouge.compute(predictions=preds, references=refs)
10
11     return {
12         "sacrebleu": float(bleu_res["score"]),
```

```

11     "rouge1": float(rouge_res["rouge1"]),
12     "rouge2": float(rouge_res["rouge2"]),
13     "rougeL": float(rouge_res["rougeL"]),
14     "rougeLsum": float(rouge_res["rougeLsum"]),
15     "n": len(preds),
16 }
17
18
19 def save_samples_md(path: Path, rows: List[Tuple[str, str, str]],
20     limit: int = 50) -> None:
21     path.parent.mkdir(parents=True, exist_ok=True)
22     lines = ["# Samples (reference vs prediction)\n"]
23     for i, (img, ref, pred) in enumerate(rows[:limit], start=1):
24         lines.append(f"## {i}\n")
25         lines.append(f"- image_path: '{img}'\n")
26         lines.append(f"- reference: {ref}\n")
27         lines.append(f"- prediction: {pred}\n")
28         lines.append("\n")
29     path.write_text("\n".join(lines), encoding="utf-8")

```

2: Baseline evaluation: automatic metrics and sample export

4 Plan of Further Work

Next up: extend the end-to-end pipeline and consolidate the baseline results obtained so far. This includes more systematic experiments, comparison of prompt variants, and deeper error analysis. We will then focus on improving grounding reliability through prompt engineering and parameter-efficient fine-tuning. Finally, we plan to conduct a more structured evaluation and package the prototype into a small demo service.

References

- [1] A. Sharma, H. Singh, M. Pant. Pixels to prose: A comprehensive survey of image captioning techniques with deep learning and generative artificial intelligence. *Neurocomputing*, vol. 667, 2026. URL: <https://www.sciencedirect.com/science/article/pii/S0925231225030577> (accessed 05.12.2025).
- [2] M. Abdulgalil, O. Basir. Next-generation image captioning: A survey of methodologies and emerging challenges from transformers to multimodal large language models. *Information Fusion*, 2025. URL: <https://www.sciencedirect.com/science/article/pii/S2949719125000354> (accessed 05.12.2025).
- [3] H. Parvin, A.R. Naghsh-Nilchi, H.M. Mohammadi. Transformer-based local-global guidance for image captioning. *Expert Systems with Applications*, 2023. URL: <https://www.sciencedirect.com/science/article/pii/S0957417423002750> (accessed 05.12.2025).
- [4] R. Wang, Y. Wu, Z. Sheng. ClipCap++: An efficient image captioning approach via image encoder optimization and LLM fine-tuning. *Expert Systems with Applications*, 2025. URL: <https://www.sciencedirect.com/science/article/pii/S156849462500780X> (accessed 05.12.2025).
- [5] S. Elbedwehy, T. Medhat, T. Hamza, M.F. Alrahmawy. Efficient image captioning based on vision transformer models. *Computers, Materials & Continua*, 2022. URL: <https://www.sciencedirect.com/org/science/article/pii/S1546221822017428> (accessed 05.12.2025).
- [6] Y. Jung, I. Cho, S.-H. Hsu, M. Golparvar-Fard. VisualSiteDiary: A detector-free vision-language transformer model for captioning photologs and retrieval. *Automation in Construction*, 2024. URL: <https://www.sciencedirect.com/science/article/pii/S092658052400219X> (accessed 05.12.2025).
- [7] A. Memiş, S. Yıldız. Tiny TR-CAP: A novel small-scale benchmark dataset for general-purpose image captioning. *Engineering Science and Technology, an International Journal*, 2025. URL: <https://www.sciencedirect.com/science/article/pii/S2215098625000643> (accessed 05.12.2025).
- [8] A. Balachandran, M.R. Masum. A multimodal framework for enhancing e-commerce information management using vision transformers and large language models. *International Journal of Information Management Data Insights*, 2025. URL: <https://www.sciencedirect.com/science/article/pii/S2667096825000370> (accessed 05.12.2025).
- [9] L. Chen, K. Li. Dual-adaptive interactive transformer for image captioning. *Pattern Recognition*, 2024. URL: <https://www.sciencedirect.com/science/article/pii/S0167865524003030> (accessed 05.12.2025).
- [10] Y. Song, Y. Wang, L. Shi, J. Yu, F. Li, S. Xiang. Token attention and attribute prediction for image captioning. *Expert Systems with Applications*, 2023. URL: <https://www.sciencedirect.com/science/article/pii/S0957417423034577> (accessed 05.12.2025).
- [11] A. Thobhani et al. Visual attention-based refined visual features for image captioning. *Computers, Materials & Continua*, 2025. URL: <https://www.sciencedirect.com/org/science/article/pii/S154622182500236X> (accessed 05.12.2025).

[12] W. Li, X. Li, Z. Li, J. Yu, P. Chen. HI-Captioner: End-to-end image captioning based on hierarchical multi-scale encoding and cross-modal interactive decoding. *Neurocomputing*, 2025. URL: <https://www.sciencedirect.com/science/article/pii/S0925231225012949> (accessed 05.12.2025).